



Feature Engineering & the Curse of Dimensionality

Building Better Inputs for Better Models

AWS Machine Learning Associate • Part 8

Presented by Prajan Shrestha

What We'll Cover



What is feature engineering?



Common techniques



What is the curse of dimensionality?



A simple analogy



Why it's a problem



How they connect & how to fight it

What is Feature Engineering?



Using domain knowledge to create, transform, or select the input variables — features — fed into a model.

Feature quality often matters more than which algorithm you choose. It's the classic "garbage in, garbage out" problem — a great algorithm fed poor features still produces a poor model.

Common Feature Engineering Techniques

Feature Creation



Deriving new features from existing ones, like pulling day-of-week from a timestamp

Transformation



Scaling or normalizing numeric values onto a comparable range

Encoding



Converting categories like colors or cities into a numeric form

Feature Selection



Deciding which features are worth keeping vs. just noise

What is the Curse of Dimensionality?



Each feature you add is another "dimension" in your dataset.

As the number of features grows, the volume of the space your data lives in grows exponentially, and your data ends up spread incredibly thin across that space.

A Simple Analogy



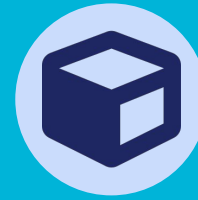
1 Dimension

A coin on a 10-ft line — easy to find



2 Dimensions

A coin in a 10×10-ft square — much harder



3 Dimensions

A coin in a 10×10×10-ft cube — harder still

The search space grows exponentially — but your data (the searchers) stays the same size, covering less and less of it.

Why It's a Problem



Data becomes sparse

You'd need exponentially more data for the same statistical coverage



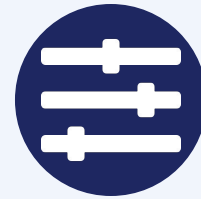
Distance loses meaning

Points that should look different start to look equally far apart



Overfitting risk rises

A model can find patterns in pure noise



Cost goes up

More dimensions mean more computation & longer training

How They Connect



Feature engineering is genuinely valuable — but it's tempting to keep adding more features, hoping more information helps.

Past a certain point, it doesn't. It actively hurts, because you run straight into the curse of dimensionality.

The goal isn't the most features — it's the most useful features.

Fighting the Curse: Solutions



Feature selection

Remove irrelevant/redundant features using correlation or importance scores



Dimensionality reduction

Techniques like PCA compress features while keeping key information



Regularization

L1/L2 penalties discourage relying on less useful features



More data

Helps offset sparsity, though the amount needed grows fast

Recap

- Feature engineering shapes raw data into useful model inputs.
- Too many features triggers the curse of dimensionality.
- Selection, reduction & regularization keep it under control.

Thanks for watching!